

		Since, E and R are constant, an <u>increase in r results in a decrease in $p.d.$</u>
21	B	Let current through $70\ \Omega$ resistor branch be I_1 Let current through the $50\ \Omega$, $30\ \Omega$ and $60\ \Omega$ resistors branch be I_2 $R_{70} // (R_{50} + R_{30} + R_{60})$ So $V_{70} = V_{50,30,60}$ $I_1 R_{70} = I_2 (R_{50} + R_{30} + R_{60})$ $70 I_1 = 140 I_2$ $I_1 = 2 I_2 \text{ ----- (1)}$ But $I_1 + I_2 = 1.5 \text{ ----- (2)}$ Solving (1) and (2), $I_2 = 0.50\text{ A}$
22	C	Use right hand grip rule at O